Ohm’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect voltage, current and resistance for an ohmic component.

SUCCESS CRITERIA - I can...

1. Identify the voltage across a resistor.
2. Use V=IR with appropriate units.
3. Predict changes while stating what stays fixed.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ohmic” and “resistance” to explain your answer. Sentence starters are optional.

ohmic: having constant resistance under the stated physical conditions

resistance: voltage divided by current for a stated operating condition

voltage: electric potential difference; energy transferred per charge

01 Warm-up: bring back an earlier idea

An electric field of **5 N/C** points left. **Draw the force direction** on **+3 C** and on **-3 C**, then **compare sizes**.

Word help: electric field = force per positive test charge at a location

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: V = IR; I = V/R.
Ohmic resistor with temperature and other physical properties kept the same. Resistance can change with temperature.

Worked reasoning: Across a **4 ohm** resistor, a **12 V** supply gives I=V/R=**3 A**. Doubling resistance at the same voltage halves current.

Your turn: Draw a one-resistor circuit, label supply voltage and current, and **mark both resistor terminals**.

Sentence starter: In my work, I used … to show … . I labeled it … .



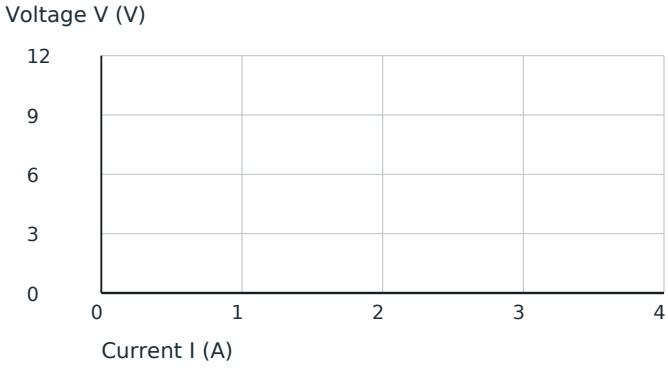
Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch a graph of V** against I for a fixed ohmic resistor.
2. What does its slope represent?

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **6 V** lab supply is connected across a **3 Ω** resistor.
1. **Label the closed circuit** and **calculate current**.
2. **Show the conventional-current direction through** the resistor.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student doubles resistance while keeping voltage fixed and predicts twice the current. Correct the prediction.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two EM waves in vacuum have frequencies f and **4** f.
Compare speeds and wavelengths.
Label a wave sketch to **support the calculation or comparison**.

Word help: vacuum = space with negligible matter

Sentence starter: Both … . They differ in … because … .



Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

Find voltage across a 4 Ω resistor carrying **2 A**.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Find three voltage-resistance pairs giving 2 A**.
2. Which comparison isolates the effect of resistance?

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3